

MuCool implementation of low energy muons in helium

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1 Why low energy physics was needed

Geant-4 is aimed at high energy simulations - MeV, GeV region - and so it does not include accurate low energy physics for our interest - keV to eV range. MuCool is a physics list made to add some of the low energy physics for positive muons in helium. This is necessary because in standard Geant-4 the following physics processes are responsible for energy loss :

- G4MuIonisation, this handles the muon energy loss due to ionisation or excitation from interactions with target electrons. This uses the stopping power from NIST tables down to 1 keV. Below 1 keV to 10 eV it uses the free electron gas model.
- G4MuMultipleScattering, this handles the change in muon direction from elastic collisions with target atoms, with no elastic energy loss.

When muon energy drops below 1 keV Geant-4 uses the model of a free electron gas, which assumes that muon energy loss is proportional to its velocity. This assumption is incorrect for positive muons moving through helium, as shown by the comparison of the stopping curve generated by standard Geant-4 and SRIM in figure 1. This is because at low energy there is not enough energy to excite or ionise target atoms (threshold effect). The two curves agree above 1 keV to approximately 200 keV because they use the same proton velocity scaled data. Whereas above 200 keV they start to disagree because Geant-4 uses Bethe-Bloch formula to predict stopping power. Then below 1 keV SRIM continues to use energy scaled proton data but with an interatomic potential. This disagreement in the stopping curves shows that Geant-4 is not able to accurately simulate the muons in helium below 1 keV, to do this muCool used proton cross section data and appropriate scaling.

2 Muon-helium collisions

There are two types of collisions - elastic and inelastic - which are classified by whether a muon interacts with helium atom as a whole, the outer electron or the nucleus. Elastic collisions change the projectile direction and energy but the internal state of the projectile and target are unchanged. Inelastic collisions change the internal state of the projectile and/or target as well. The main inelastic collisions include : excitation, ionisation and electron capture. An additional process is the re-ionisation of muonium, but muonium may lose energy elastically or through other inelastic processes.

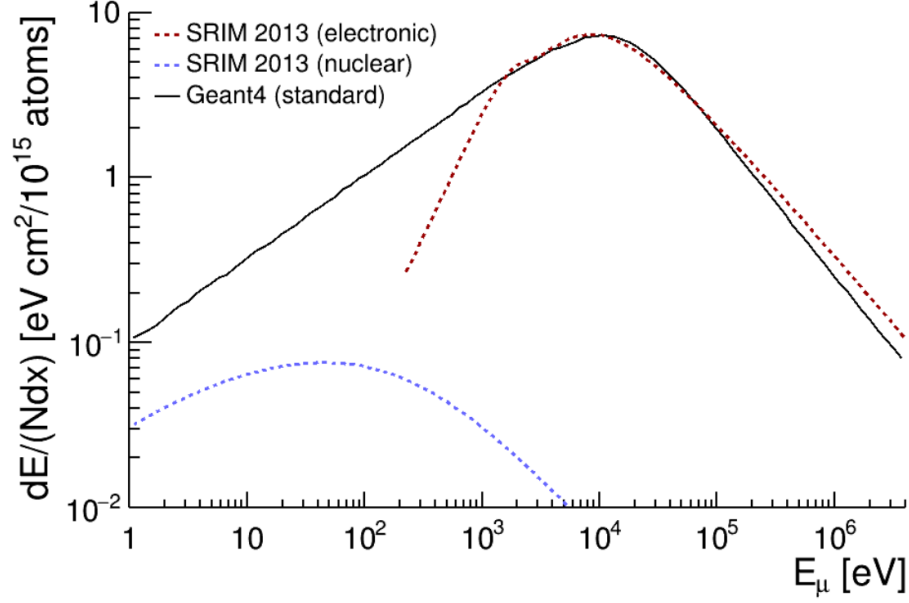


Figure 1: Stopping curve of positive muons in helium generated in standard Geant-4 with no low energy physics and SRIM. To show that below 1 keV standard Geant-4 is not appropriate. Taken from [?]]

3 What muCool actually does

Above 1 keV standard Geant-4 is used, so only below 1 keV is muCool used. Below 1 keV elastic collisions and charge exchange processes are implemented. These processes are carried out in steps, at each step one of the competing processes is chosen and carried out. Each collision is done one at a time and between collisions the muon (muonium) propagates freely.

- The mean free path of a collision type, i , is given by equation 1 where N is the gas density (He atoms per cm^3), $\sigma_i(E_\mu)$ is the cross section for the collision type (electron loss or capture, or elastic)
- The probability that the muon undergoes a collision can then be calculated using equation 2 where l_i is the distance a muon has travelled between two collisions of the same type, i .
- Due to three competing processes the probability has to be generalised to equation 4 this is because the number of mean free paths, $n_{\lambda,i}$, is energy independent but this is not fulfilled in equation 2. This ensures that on average a muon will travel one mean free path between two collisions of the same type.

$$\lambda_i(E_\mu) = \frac{1}{N\sigma_i(E_\mu)} \quad (1)$$

$$P_i(l_i) = 1 - e^{-\frac{l_i}{\lambda_i}} \quad (2)$$

$$n_{\lambda,i} = \sum_j \frac{\Delta l_j}{\lambda_{i,j}} \quad (3)$$

$$P_i = 1 - e^{-n_{\lambda,i}} \quad (4)$$

- At first each muon is given a randomly generated, $n_{\lambda,i}$, according to the exponential distribution, $e^{-n_{\lambda,i}}$, that has a mean value of one
- Each collision type has a step length given by equation 5. The process with the smallest step length is the process that is chosen.

$$\Delta l_i = n_{\lambda,i} \lambda_i \quad (5)$$

- For the next step the process that was carried out has its number of mean free paths updated by using the exponential distribution. The two processes that were not carried out are updated using equation 6 where Δl_i is the length of the previous step defined by the chosen process
- Each process proposes a new step length and the shortest is picked. This process is repeated until the end of the simulation

$$n_{\lambda,j} = n_{\lambda,j} - \frac{\Delta l_i}{\lambda_i} \quad (6)$$

3.1 If an elastic collision is chosen

When low energy elastic collision is picked the muon (muonium) direction and kinetic energy are changed at the end of the step. This is done by carrying out below :

- The deflection angle, χ , is randomly generated in the centre of mass frame using the $\cos\chi$ uniform distribution between -1 and 1.
- The elastic energy loss is calculated using equation 7 where E_μ is the muon kinetic energy in lab frame.
- Muon energy is updated : $E'_\mu = E_\mu - dT$

$$dT = \frac{2m_\mu m_{He}}{(m_\mu + m_{He})^2} (1 - \cos\theta) E_\mu \quad (7)$$

- Next the direction needs to be changed
- First the centre of mass scattering angle χ is transformed into the scattering angle, θ , in the lab frame using equation 8
- Using both θ and additional azimuthal angle, ϕ , generated uniformly between 0 and 2π the muon momentum is rotated

$$\tan\theta = \frac{\sin\chi}{\cos\chi + \frac{m_\mu}{m_{He}}} \quad (8)$$

3.2 If an inelastic collision is chosen

Instead if electron capture or loss is chosen then the following is carried out :

- The kinetic energy and charge are modified but direction is unaffected
- After electron capture charge is changed to zero
- After electron loss charge is changed to one
- Kinetic energy loss is determined by equations 9, 10 and 11

$$Q_{loss} = 13.6 \text{ eV} + \Delta Q \quad (9)$$

$$Q_{capture} = 11 \text{ eV} + \Delta Q \quad (10)$$

$$\Delta Q = \begin{cases} 220 \text{ MeV cm}^2 / g \cdot \rho \cdot l \cdot \frac{E_\mu - 300 \text{ eV}}{700 \text{ eV}} & \text{for } E_\mu > 300 \text{ eV} \\ 0 & \text{for } E_\mu < 300 \text{ eV} \end{cases} \quad (11)$$

4 Where the equations came from

4.1 Scaling from proton data

To get the cross sections for muon interactions the proton cross section was scaled as shown in figure 2. The scaling is either velocity or energy scaled and this depends on the process. For momentum transfer, elastic, it was energy scaled (using equation 12) and for electron loss and capture it was velocity scaled. This means that electronic stopping power is the same for muons and protons at the same velocity, and nuclear stopping power is the same at the same centre of mass energy. From figure 2 can see that below 100 eV that only elastic processes occur and all inelastic processes are suppressed. Between a few hundred eV and 10 keV inelastic process dominate, above approximately 800 eV all elastic processes are suppressed. This is the cross section data that was used to recreate the SRIM stopping curve.

$$S_{n,p}(E_{cm}) = 7.3 \cdot S_{n,\mu}(E_{cm}) \quad (12)$$

4.2 Probability and outcome of collisions

To model the low energy interactions the probability and outcome of the collisions was necessary. To calculate this, the total cross section was needed shown in equation 13, where the projectile energy is E , Ω is the solid angle and ΔE is the energy loss.

$$\sigma(E) = \int \int \frac{d^2\sigma}{d(\Delta E) d\Omega}(\Omega, \Delta E, E) d(\Delta E) d\Omega \quad (13)$$

The total probability is proportional to the cross section shown in 13. The total cross section can be simplified to equation 14 because energy loss of particles is one-to-one function of the deflection angle so can be described by either $\frac{d\sigma}{d\Omega}(\Omega, E)$ or $\frac{d\sigma}{d(\Delta E)}(\Delta E, E)$

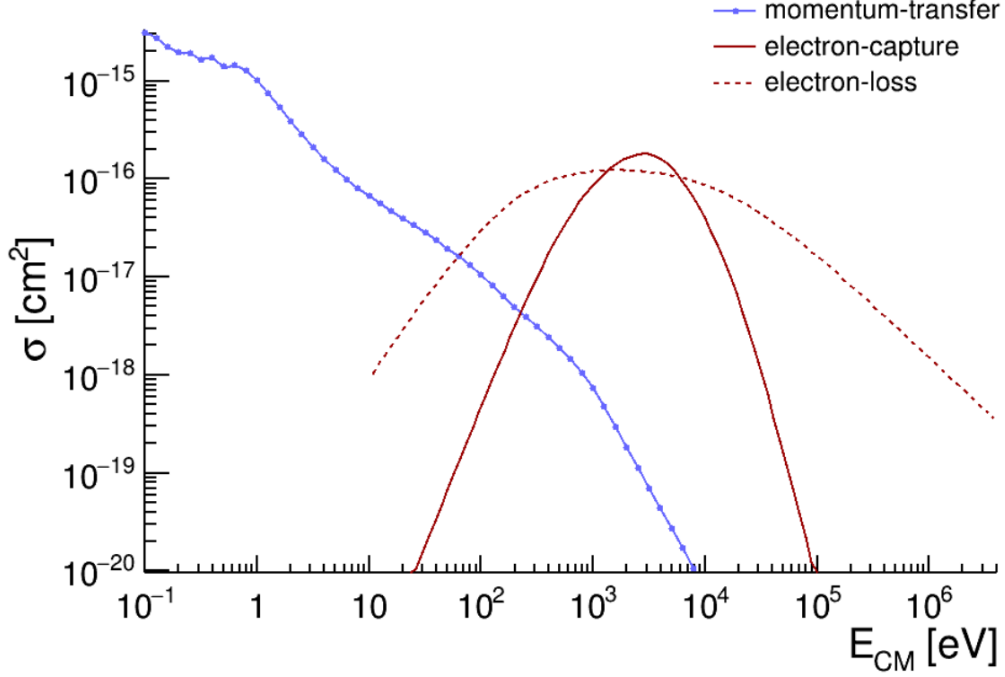


Figure 2: Muon cross sections for charge exchange and elastic collisions scaled from proton cross section data. Momentum transfer cross section energy scaled by equation 12 and charge exchange processes by velocity scaling. Figure taken from [?]

$$\sigma(E) = \int \frac{d\sigma}{d\Omega}(\Omega, E) d\Omega = \int \frac{d\sigma}{d(\Delta E)}(\Delta E, E) d(\Delta E) \quad (14)$$

To calculate an approximation for the stopping power equation 15 is used

$$S(E) = \overline{\Delta E} \cdot \sigma(E) \quad (15)$$

4.3 Elastic collisions

The elastic stopping power used in the muCool code is shown in equation 16, using the momentum transfer cross section shown in figure 2.

$$S_{el}(E_{cm}) = \frac{2m_A}{m_A + m_B} E_{cm} \sigma_{MT}(E_{cm}) \quad (16)$$

Where the momentum transfer cross section, σ_{MT} , is shown in equation 17. Described by the differential cross section I_{el} at large deflection angles χ which are what produce large energy losses.

$$\sigma_{mt}(E_{cm}) = 2\pi \int_0^\pi I_{el}(\chi, E_{cm}) (1 - \cos\chi) \sin\chi d\chi \quad (17)$$

But using conservation of energy and momentum it can be found that the energy loss in the lab frame is given by equation 7. From this equation can see that energy loss is determined by the scattering angle χ with backward scattering being the maximum possible energy loss given by equation 18.

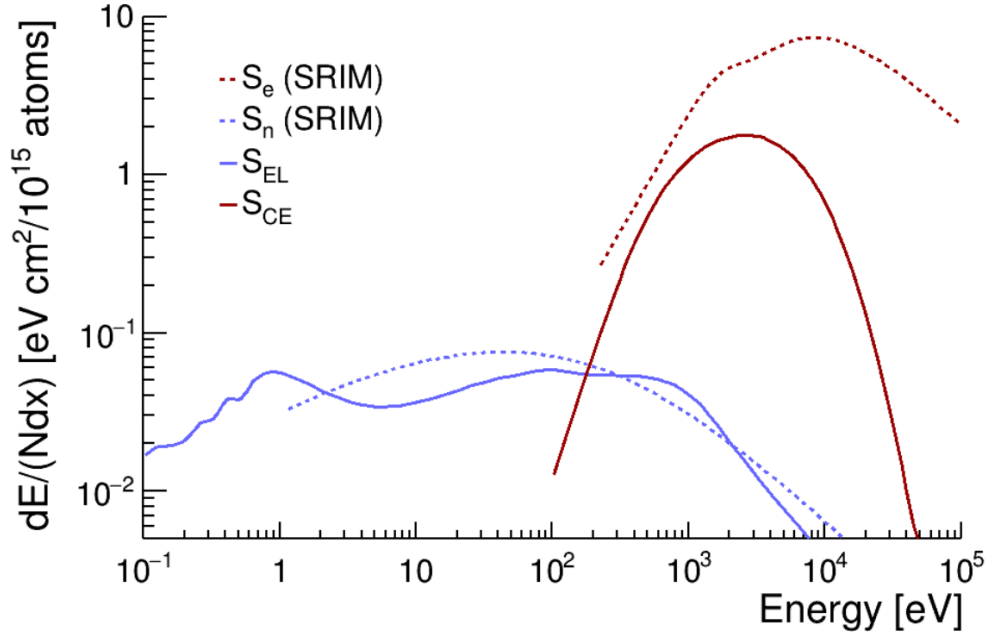


Figure 3: Stopping powers calculated using scaled proton cross section data compared to stopping power generated by SRIM. Electronic stopping powers not matching because missing inelastic processes in the proton cross section data. Nuclear not matching because SRIM uses an interatomic potential but muCool used a Born-Oppenheimer approximation. Figure taken from [?]

$$dT_{max} = \frac{4m_{\mu}m_{He}}{(m_{\mu} + m_{He})^2}E_{\mu} \quad (18)$$

The shape of the nuclear stopping powers shown in figure 3 are different because of the potential used in the calculation. In SRIM they use an interatomic potential whereas muCool used the more accurate Born-Oppenheimer approximation.

4.4 Inelastic collisions

The stopping power for inelastic processes can be calculated by 19 where Q is the internal energy loss.

$$S_{Inel} = Q \cdot \sigma_{Inel}(E_{cm}) \quad (19)$$

Above 100 eV inelastic processes are the dominant energy loss mechanism, specifically charge exchange. In this range the muon goes through a series of charge exchange cycles and so spends time as muonium. The time spent as muonium or as muon is depends on equations 20 and 21. Whichever cross section, loss or capture, is bigger determines whether there is more muonium or muons.

$$f_{Mu}(E_{cm}) = \frac{\sigma_{capture}(E_{cm})}{\sigma_{loss}(E_{cm}) + \sigma_{capture}(E_{cm})} \quad (20)$$

$$f_{\mu^+}(E_{cm}) = \frac{\sigma_{loss}(E_{cm})}{\sigma_{loss}(E_{cm}) + \sigma_{capture}(E_{cm})} \quad (21)$$

At energies less than 1 keV the electron loss cross section is much larger than electron capture, this means any muonium formed at this energy is likely to be reionised. This reduces the loss of muons to muonium formation. This means that muons slowing down and stopping in helium gas spend a significant amount of time as muonium in keV energies, but slowed to lower eV ranges this muonium is likely reionised.

$$S_{capture}(E_{cm}) = 11 \text{ eV} \cdot \sigma_{capture}(E_{cm}) \quad (22)$$

$$S_{loss}(E_{cm}) = 13.6 \text{ eV} \cdot \sigma_{loss}(E_{cm}) \quad (23)$$

To calculate the total stopping power from both processes the fractions shown in 21 and 20 need to be considered

$$\begin{aligned} S_{cap+loss} &= f_{\mu^+}(E_{cm}) S_{capture}(E_{cm}) + f_{Mu}(E_{cm}) S_{loss}(E_{cm}) \\ &= 24.6 \text{ eV} \cdot \frac{\sigma_{loss}(E_{cm}) \sigma_{capture}(E_{cm})}{\sigma_{loss}(E_{cm}) + \sigma_{capture}(E_{cm})} \end{aligned} \quad (24)$$

Equation 24 is what is plotted in figure 3, this stopping accounts for the energy loss from electron capture and loss. This stopping power is smaller than the SRIM calculation because electron capture and loss are not the only mechanisms for energy loss in this energy range. There are other inelastic processes that have not been accounted for in equation 24, such as muonium excitation. Muonium excitation is the largest missing contribution in this energy range with the cross section being similar to electron loss. There are other contributing processes such as helium excitation and ionisation but these are one to two orders of magnitude smaller than the other inelastic cross sections between 100 eV to 1 keV so do not contribute much to the stopping power.

Instead of implementing these processes completely, the muCool code adds an additional inelastic energy loss as shown in equation 11 to match the SRIM stopping curve. This additional energy loss ranged from on average 12 eV at 1 keV to 0 eV at 300 eV. This was done because for their purpose of the code the muons did not spend much time in this energy range as they slowed the muons to between 1 eV and 100 eV. This would only impact their initial slowing down process, therefore altering only the initial muon stopping distribution in helium without impacting the later muon drift. This is because below 100 eV is handled by elastic collisions which is implemented completely. The cutoff is 300 eV for the additional energy loss because there isn't any cross section below 300 eV for these extra processes, and so it is not clear how the other inelastic process behave below 300 eV.

Another addition made to the code is to include muonium elastic scattering. This doesn't impact total stopping power a lot but could impact how many muons survive. If there is muonium with energy just above 13.6 eV then if it scatters elastically the energy may drop below this threshold. This would mean that electron loss process would no longer be possible and so the muon would remain bound until it decays. There are no theoretical predictions for muonium and helium momentum transfer and so it was assumed the cross section was the same as muon helium momentum transfer.

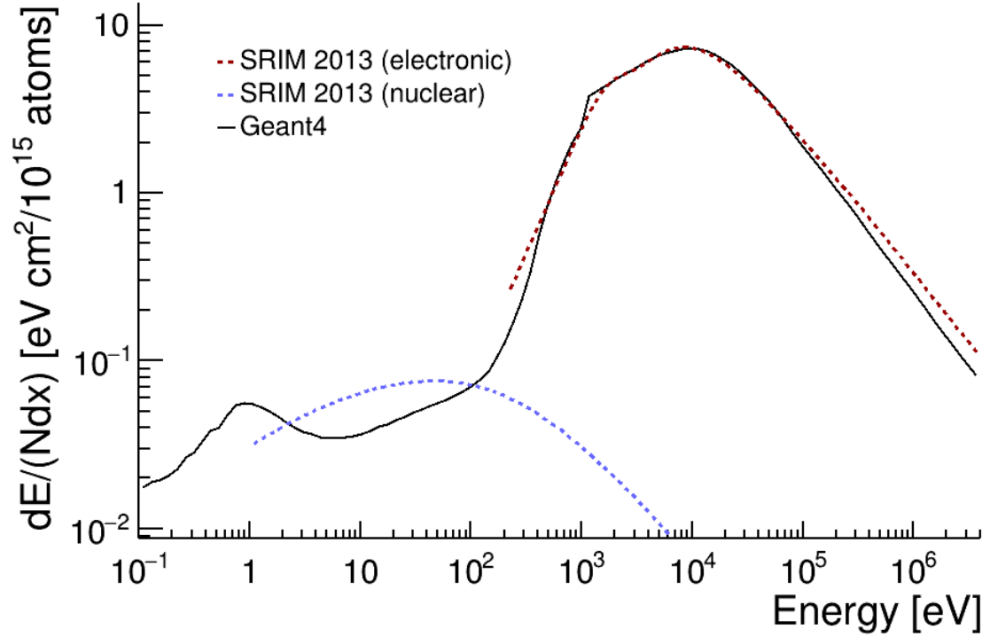


Figure 4: Stopping power curve for muons in helium with the extra inelastic energy loss so that electronic stopping power from SRIM matches Geant-4. Figure taken from [?]

5 Limitations and summary of energy cuts

- MuCool adds extra physics processes : electron loss, capture and elastic collisions. These are meant to be turned for muon energy less than 1 keV
- Between a few hundred eV and 1 keV charge exchange dominants
- Below 100 eV it is only elastic collisions that is turned on
- Other inelastic processes such as muonium excitation were not implemented properly and instead an additional inelastic energy loss is added between 1 keV and 300 eV. This did not affect muCool simulations because muons did not spend much time above 100 eV.
- The SRIM and muCool stopping curves match after adding the low energy physics, the only differences are the shapes of the nuclear sections because of the potential each uses when calculating. And above 200 keV Geant-4 uses Bethe-Bloch and SRIM uses proton data.

References